



Pranesh S A 251801234-2025-AIDS ▾

P2

Started on	Thursday, 24 September 2026, 8:34 AM
State	Finished
Completed on	Thursday, 24 September 2026, 9:27 AM
Time taken	52 mins 30 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b
s2		g	x	t	x	a	y b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s1[100];
6      char s2[100];
7      scanf("%s", s1);
8      scanf("%s", s2);
9
10     int m = strlen(s1);
11     int n = strlen(s2);
12
13     int dp[101][101];
14
15     int i = 0;
16     while (i <= m) {
17         int j = 0;
18         while (j <= n) {
19             if (i == 0 || j == 0) {
20                 dp[i][j] = 0;
21             } else if (s1[i - 1] == s2[j - 1]) {
22                 dp[i][j] = dp[i - 1][j - 1] + 1;
23             } else {
24                 int val1 = dp[i - 1][j];
25                 int val2 = dp[i][j - 1];
26                 if (val1 > val2) {
27                     dp[i][j] = val1;
28                 } else {
29                     dp[i][j] = val2;
30                 }
31             }
32             j++;
33         }
34         i++;
35     }
36 }
```

```
36  
37     printf("%d\n", dp[m][n]);  
38  
39     return 0;  
40 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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